Fairness-aware Analysis of Centrality Measures in Social Networks

Kirtidev Mohapatra
kirtidevm@iitbhilai.ac.in
IIT Bhilai

Rishi Ranjan Singh
rishi@iitbhilai.ac.in
IIT Bhilai

Akrati Saxena
a.saxena@liacs.leidenuniv.nl
Leiden University

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Introduction

It is essential to ensure fairness in Influence spreading among groups in a network to ensure equal access to resources and information for each group. Uneven information sharing can generate disparity among the nodes in the network, thus depriving them of essential benefits. Influence Maximization is an NP-Hard problem, and identifying the potential set of seed nodes to ensure fairness is a point of contention. Analysing centrality measures to identify potential seed nodes is essential for reducing inequality in information spreading among different groups within a network. This helps balance the polarity in the information access within groups in the network.

Problem Statement

Identify centrality measure for fair influence spreading among communities of network.

Methodology

- Homophily BA model is used for generating the network.
- Centrality measures are used to select seed nodes.
- The Independent Cascading Model (ICM) is used to spread influence.
- Fairness metrics are used to compare the centrality measures.

Centrality Measures

Local Centralities focuses on a node's immediate neighbour. It is least bothered about the entire structure of the network, such as Degree, Clustering, H-index, and H2-index.

Global Centralities computes a node's importance by considering the entire network structure. These centralities are more computationally expensive than local centralities. Global centralities include Betweenness, Closeness, Page Rank, Katz, Harmonic, LFPR(Local Fair Page Rank) and FSPR(Fair Sensitive Page Rank).

Semi-local Centrality is a tradeoff between local centrality and global centrality. Its purpose is to rank nodes similar to global centrality at the same time, the computational cost to be reduced as of local centrality. Some examples are LSC(Locally Structural Centrality), LASPN(Local Average Shortest Path with Extended Neighbourhood Concept), and Duanbing.

Modular centrality computes the node's importance within a particular community or subgraph. Some examples of modular centrality are Participation Coefficient & Bridging.

Fairness Metrics

L1- NORM

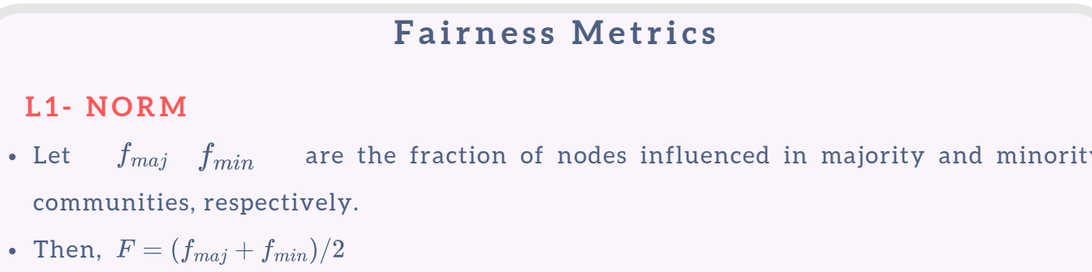
- Let f_{maj} f_{min} are the fraction of nodes influenced in majority and minority communities, respectively.
- Then, $F = (f_{maj} + f_{min})/2$
- L1 norm is given as : $|F - f_{maj}| + |F - f_{min}|$

DISPARITY

- It is the ratio of the fraction of nodes influenced in community 1 to the fraction of nodes influenced in community 2.
- Disparity can be given by: $Disp\,(centrality) = f_{maj}/f_{min}$

Experimental Setup

- Minority nodes are less prominent with more homophilic networks.
- A total population of 10,000 nodes is being used.
- Minority is 0.2% of the total population, and the majority is 0.8% of the total population.
- 100 simulations are performed.



Results

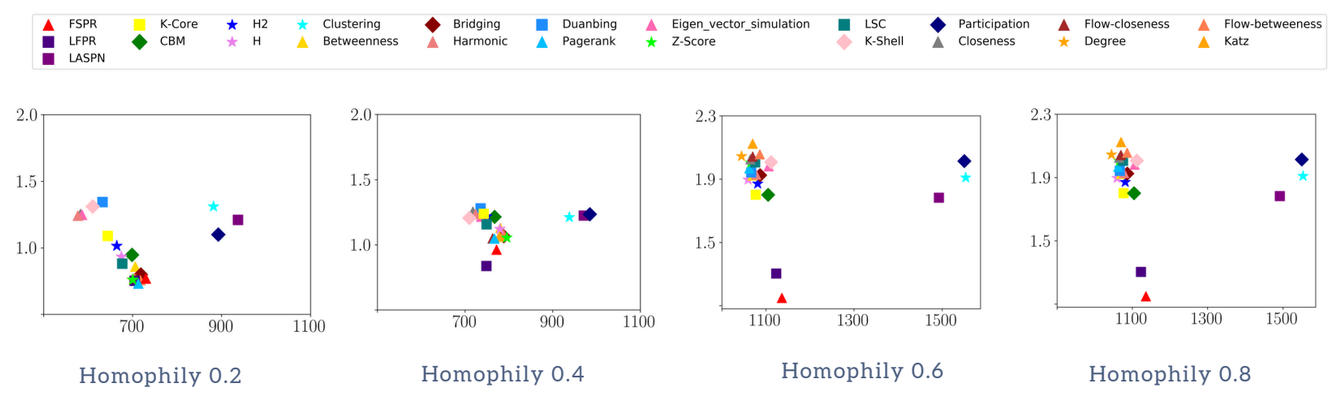


Figure 1 represents different results obtained from different homophily values starting from the left, With a seed node size of 1% of the total population. The y-axis denotes the Disparity computed values, and the x-axis represents the number of influenced nodes.

- Lower disparity scores indicate the centrality of being fair.
- It is observed that LFPR and FSPR are less biased compared to other centralities..

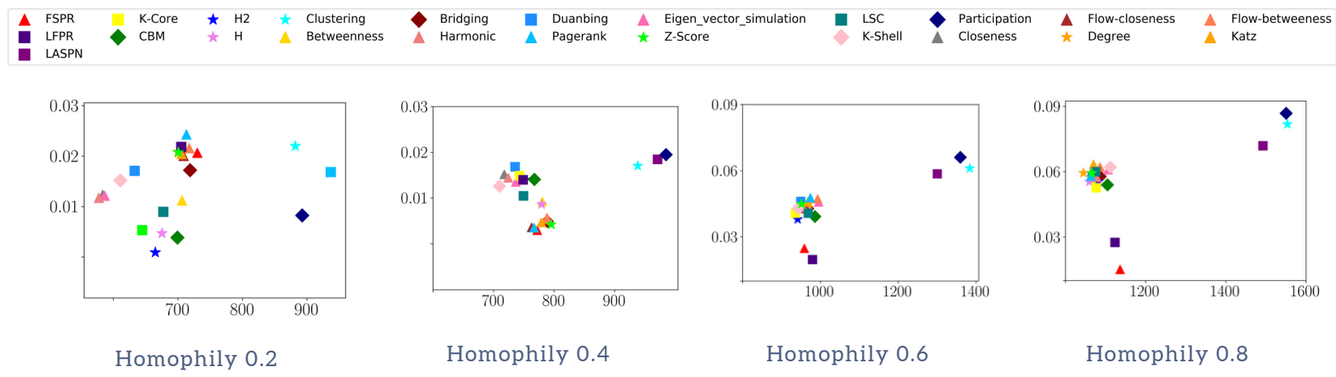


Figure 2 represents different results obtained from different homophily values starting from the left, With a seed node size of 1% of the total population. The y-axis denotes the L1 Norm computed values, and the x-axis represents the number of influenced nodes.

- The lower L1 norm indicates that the centrality is fair.
- LFPR and FSPR have lower L1 norm scores.
- In homophilic networks, LFPR and FSPR consistently spread influence among communities.

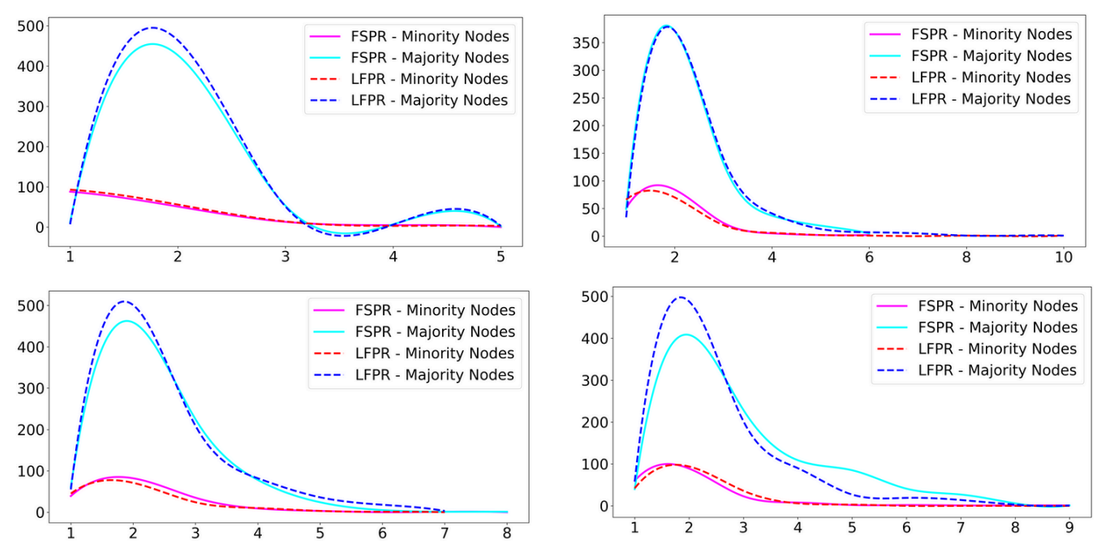


Figure 3 represents detailed comparison of LFPR and FSPR, where x- axis represents number of iterations required to converge and the y axis represents the number of nodes influenced. starting from left each plot represents Homophily values from 0.2 to 0.8

Conclusion

- FSPR [3] and LFPR [3] were fairer, as observed in Fig 1 and Fig 2.
- LFPR enforces fairness at the node level.
- FSPR enforces fairness at the global level.
- LFPR and FSPR controlled the random walk to ensure a fair share of community ranking.

Reference

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[2] Saxena, A., Fletcher, G., & Pechenizkiy, M. (2024). Fairsna: Algorithmic fairness in social network analysis. ACM Computing Surveys, 56(8), 1–45

[3] Tsioutsiouliklis, Sotiris, Pitoura, Evaggelia, Tsaparas, Panayiotis, Kleftakis, Ilias, and Mamoulis, Nikos. "Fairness-aware PageRank." In Proceedings of the Web Conference 2021, pages 3815-3826, 2021.